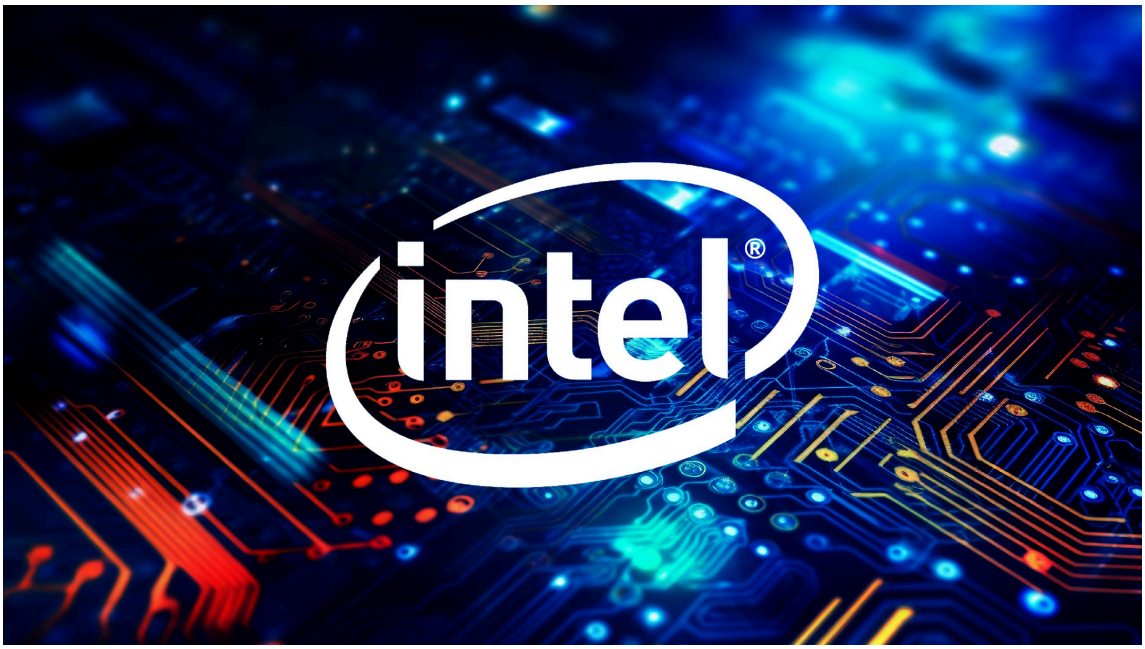
INTEL CORPORATION



iGPT

Architecture Overview

Intel Generative Pre-trained Transformer Platform

Version: 1.0

Date: March 2026

Prepared by: Sajiv Francis

Classification: Internal Use

This document contains confidential information proprietary to Intel Corporation.

Table of Contents

1. Executive Summary
2. iGPT Key Capability Patterns
3. iGPT Platform Characteristics
4. Architecture Overview
5. Sub-Architecture: Chat & Conversation Flow
6. Sub-Architecture: Assistant Builder & Lifecycle
7. Sub-Architecture: Data Management & RAG
8. Sub-Architecture: Marketplace & Sharing
9. Module Capabilities
    - Chat Interface · Assistant Builder · My Assistants · Data Management
    - Trainer · Sharing & Publishing · Marketplace · Learn & Resources
    - Model Management · Setup & Configuration
10. Security & Governance
11. External Integrations
12. Related Architecture Patterns
13. Appendix A: Component Glossary

Executive Summary

iGPT is Intel's internal enterprise Generative AI platform that provides employees with conversational AI capabilities, custom assistant creation, and a marketplace ecosystem for sharing AI solutions across the organisation.

- **Chat Interface** – Central conversational AI for multi-turn interactions with enterprise LLMs (GPT-4, GPT-3.5, and other models)
- **Assistant Builder** – No-code wizard for creating, configuring, and publishing custom AI assistants with model selection, system prompts, temperature control, and live testing
- **Marketplace** – Enterprise-wide assistant store for discovering, sharing, and reusing AI assistants across Intel

Key Architectural Components:

- **iGPT Web Application** – Primary entry point at igpt.intel.com/chat; responsive SPA with sidebar navigation, chat workspace, and builder panels
- **LLM Gateway** – Model routing layer supporting GPT-4, GPT-3.5, and other enterprise LLMs with temperature and parameter controls
- **RAG Engine** – Retrieval Augmented Generation for document-grounded responses using uploaded files and data pipelines
- **Intel SSO / Azure AD** – Enterprise authentication via Single Sign-On with Azure Active Directory integration
- **AGS Entitlements** – Fine-grained access control for Secure Group sharing of assistants
- **Marketplace Service** – Discovery, publishing, and catalog management for shared assistants

The architecture ensures secure, governed AI assistance while maintaining enterprise identity management and InfoSecurity compliance.

> **InfoSecurity Restriction:** Assistants with attached files can only be published as Private. This is a directive from InfoSecurity, not a technical limitation.

iGPT Key Capability Patterns

The iGPT platform supports four primary interaction patterns:

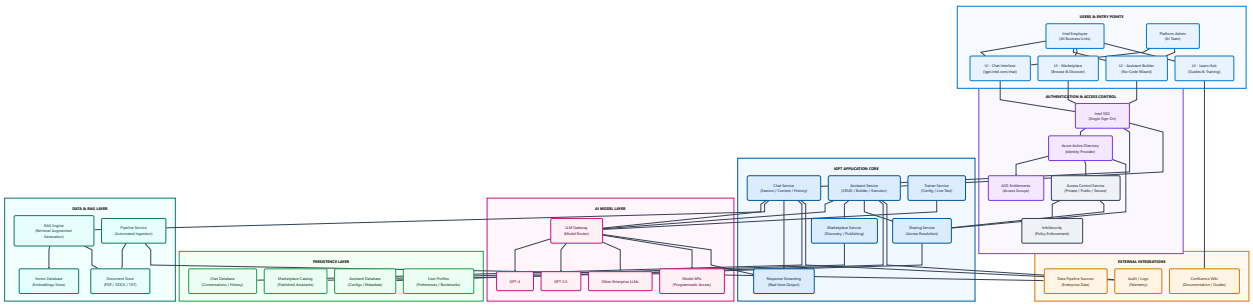
Pattern	Description
Conversational	Multi-turn chat with enterprise LLMs. Users interact via natural language to get answers, generate content, analyse data, and complete tasks. Supports model selection and streaming responses.
Assistive	Custom AI assistants with specialised system prompts, configured models, and domain-specific instructions. Built via the no-code Assistant Builder with live testing.
Collaborative	Share assistants across Intel via the Marketplace. Publish as Public (all employees), Secure Groups (Azure AD groups), or keep Private (owner-only).
Knowledge-Augmented	RAG-powered responses grounded in uploaded documents (PDF, DOCX, TXT) and automated data pipelines. Enables context-aware AI within enterprise data.

iGPT Platform Characteristics

Characteristic	Description
Enterprise Chat AI	Production-ready conversational interface with multiple LLM backends, streaming responses, and conversation history.
No-Code Assistant Creation	Step-by-step wizard for building AI assistants without programming – name, model, temperature, system prompt, test, publish.
Marketplace Ecosystem	Central store for browsing, searching, bookmarking, and running shared assistants across Intel.
Data-Augmented Intelligence	Document upload and pipeline integration enabling RAG-based retrieval for grounded, enterprise-specific responses.
Security by Default	SSO authentication, Azure AD group access, private-by-default assistants, and InfoSecurity compliance enforcement.
Model Flexibility	Multiple LLM options (GPT-4, GPT-3.5, others) with per-session model selection and temperature control via API gateway.

Architecture Overview

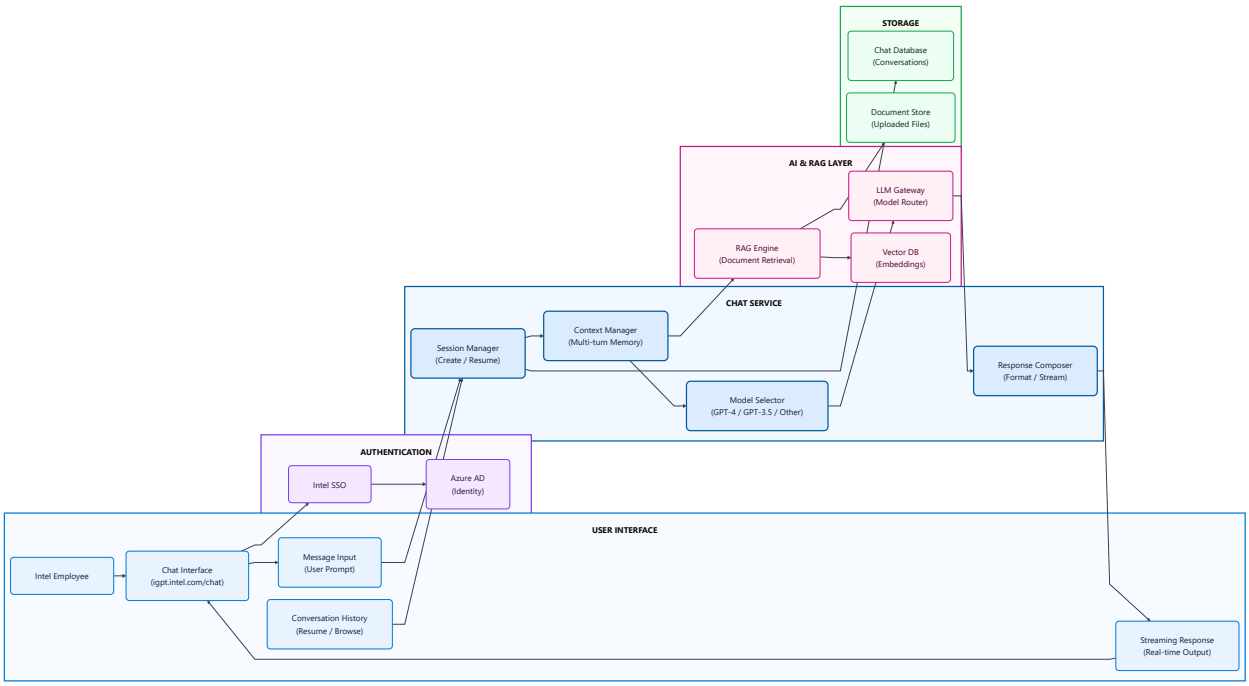
Architecture Overview – iGPT Platform Component Architecture Diagram



Sub-Architecture: Chat & Conversation Flow

Purpose: Detail the end-to-end conversational flow from user input through LLM processing to streaming response delivery.

Chat & Conversation Flow Architecture



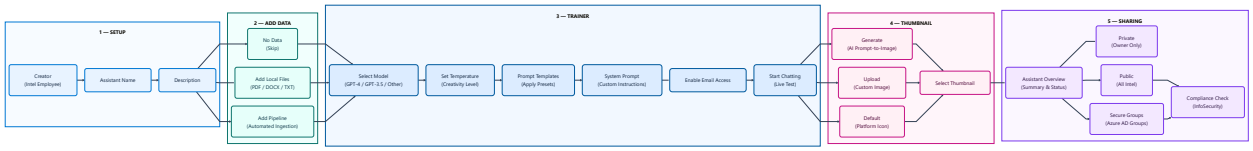
Key Components:

- **Session Manager** – Creates and resumes chat sessions; maintains session state across multiple interactions
- **Context Manager** – Preserves multi-turn conversation memory; passes relevant history to LLM for contextual responses
- **Model Selector** – Per-session model choice (GPT-4, GPT-3.5, others); configurable temperature and parameters
- **RAG Engine** – Retrieves relevant document chunks from Vector DB when assistant has attached data
- **Response Composer** – Formats LLM output and streams tokens in real-time back to the UI

Sub-Architecture: Assistant Builder & Lifecycle

Purpose: Detail the 5-tab assistant creation workflow: Setup, Add Data, Trainer, Thumbnail, and Sharing.

Assistant Builder & Lifecycle Architecture



Workflow (5-Tab Builder):

> **Setup** (Name & Description) → **Add Data** (None / Local Files / Pipeline) → **Trainer** (Model Selection, Temperature, Prompt Templates, System Prompt, Email Access, Start Chatting) → **Thumbnail** (Generate / Upload / Default → Select) → **Sharing** (Assistant Overview → Private / Public / Secure Groups)

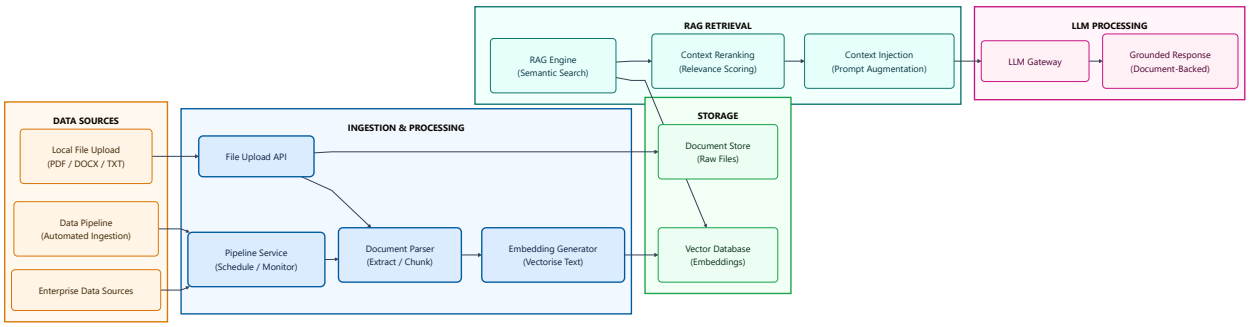
Key Design Decisions:

- **Private by Default** – All new assistants default to Private (owner-only) access
- **Compliance Gate** – Public and Secure Group publishing requires a compliance checkbox
- **File Restriction** – Assistants with attached files must remain Private (InfoSecurity directive)
- **Live Testing** – "Start Chatting" in the Trainer tab provides real-time testing against the configured LLM
- **AI Thumbnails** – Users can write a prompt to generate a thumbnail image, upload a custom image, or use the platform default

Sub-Architecture: Data Management & RAG

Purpose: Detail the data ingestion, document processing, and Retrieval Augmented Generation pipeline.

Data Management & RAG Architecture



Key Components:

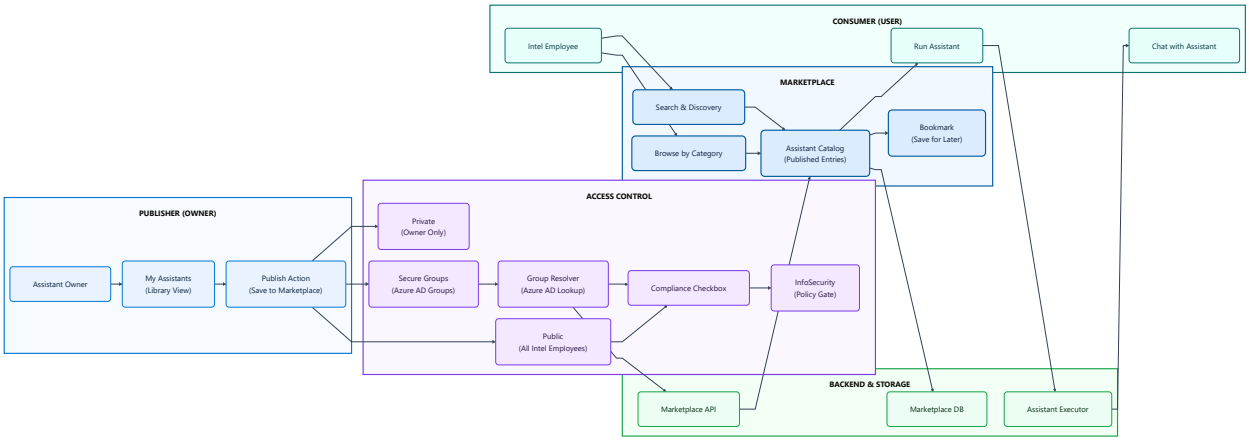
- **File Upload API** – Accepts PDF, DOCX, TXT; validates format and size constraints
- **Document Parser** – Extracts text content and chunks documents into retrieval-sized segments
- **Embedding Generator** – Converts text chunks into vector embeddings for semantic search
- **RAG Engine** – Performs semantic similarity search against Vector DB at query time
- **Context Injection** – Augments the user prompt with retrieved document context before LLM processing

Data Restriction: File-attached assistants remain Private per InfoSecurity directive.

Sub-Architecture: Marketplace & Sharing

Purpose: Detail the marketplace ecosystem and access control model for assistant sharing.

Marketplace & Sharing Architecture



Access Control Hierarchy:

> **Private** (Owner Only) → **Secure Groups** (Azure AD Groups via AGS) → **Public** (All Intel Employees)

Key Design Decisions:

- **Owner Visibility** – Owners see sharing status icons in My Assistants but do not see their own assistants in the Marketplace view
- **Azure AD Groups** – Secure Group sharing resolves membership against Azure AD; only Azure AD groups are visible in the iGPT interface
- **Compliance Gate** – Both Public and Secure Group sharing require an explicit compliance checkbox before publishing

Module Capabilities

Chat Interface

Capability	Description
Create Chat	Start new conversation sessions with selected models
Multi-turn Conversations	Maintain context across multiple exchanges
Model Selection	Choose LLM per session (GPT-4, GPT-3.5, others)
Conversation History	Browse, resume, and manage past conversations
Streaming Responses	Real-time token streaming from LLM to UI
Chat with Assistants	Use custom or marketplace assistants in conversation

Assistant Builder

The Assistant Builder follows a 5-tab workflow: **Setup** → **Add Data** → **Trainer** → **Thumbnail** → **Sharing**.

Tab 1: Setup

Capability	Description
Assistant Name	Define a unique name for the assistant
Description	Provide a summary of the assistant's purpose and capabilities

Tab 2: Add Data

Capability	Description
No Data	Skip data attachment; assistant relies solely on LLM knowledge
Add Local Files	Upload PDF, DOCX, TXT files for RAG-based retrieval
Add Pipeline	Configure automated data ingestion from enterprise data sources

Tab 3: Trainer

Capability	Description
Select Model	Choose LLM for the assistant (GPT-4, GPT-3.5, or other enterprise models)
Set Temperature	Adjust creativity level (low = deterministic, high = creative)
Prompt Templates	Apply pre-built prompt templates to guide assistant behaviour
System Prompt	Write custom instructions defining the assistant's persona and rules
Enable Email Access	Grant the assistant access to email context for responses
Start Chatting	Live test the assistant against the configured LLM before saving

Tab 4: Thumbnail

Capability	Description
Generate	Write a prompt to AI-generate a thumbnail image for the assistant
Upload	Upload a custom image as the assistant thumbnail
Default	Use the platform default icon
Select Thumbnail	Preview and select the final thumbnail from generated/uploaded options

Tab 5: Sharing

Capability	Description
Private (default)	Owner-only access; assistant is not visible to others
Public	All Intel employees can discover and run the assistant
Secure Groups	Restrict access to specific Azure AD / AGS entitlement groups
Assistant Overview	Review summary, configuration, and sharing status before publishing
Compliance Check	Required checkbox for Public and Secure Group publishing (InfoSecurity)

My Assistants

Capability	Description
Owned Assistants	View all assistants created by the user
Bookmarked	Access assistants saved from Marketplace
Edit / Delete	Manage owned assistant configurations
Chat	Start conversation with any owned assistant
Status Icons	Visual indicators: Public / Secure / Private / Bookmarked / Owned

Data Management

Capability	Description
No Data	Skip data attachment; rely on base LLM knowledge only
Add Local Files	Upload PDF, DOCX, TXT for RAG-powered retrieval within the assistant
Add Pipeline	Configure automated data ingestion pipelines from enterprise sources
File Formats	PDF, DOCX, TXT, and other document types
File Privacy	Assistants with attached files must remain Private (InfoSecurity directive)

Trainer

Capability	Description
Select Model	Choose the LLM backend for the assistant (GPT-4, GPT-3.5, or other enterprise models)
Set Temperature	Adjust creativity / determinism level for the assistant's LLM responses
Prompt Templates	Apply pre-built prompt presets to accelerate assistant configuration
System Prompt	Define custom instructions, persona, rules, and behavioural guidelines
Enable Email Access	Grant the assistant access to email context for enriched responses
Start Chatting	Live-test the assistant in real-time against the configured LLM before saving

Thumbnail

Capability	Description
Generate	Write a text prompt to AI-generate a visual thumbnail for the assistant
Upload	Upload a custom image file as the assistant thumbnail
Default	Use the platform's default assistant icon
Select Thumbnail	Preview generated / uploaded options and confirm the final selection

Sharing & Publishing

Sharing Mode	Description
Private (default)	Owner-only access and modification rights
Public	All Intel employees can run; only owner can modify
Secure Groups	Azure AD / AGS groups can run; only owner can modify
Assistant Overview	Summary view of assistant configuration, data, and sharing status before publishing
Compliance Check	Required checkbox acknowledgment before Public or Secure Group publishing

Additional: Compliance checkbox required for Public and Secure Group publishing. File-attached assistants must remain Private.

Marketplace

Capability	Description
Browse	Discover assistants published by other teams
Search & Filter	Find assistants by category and keyword
Bookmark	Save interesting assistants for later access
Run	Execute any accessible marketplace assistant
Publish	Share your assistants with the enterprise

Learn & Resources

Resource	Type
Prompt Formula Training	Video tutorial
Create iGPT Assistants	Step-by-step guide (Confluence)
Share in Marketplace	Publishing guide (Confluence)
Model Licensing	Compliance and licensing information

Model Management

Capability	Description
Model Catalog	Multiple LLM options with descriptions
Temperature Control	Per-session randomness/creativity adjustment
Model APIs	Programmatic access to models
Licensing Info	Model compliance and licensing details

Setup & Configuration

Capability	Description
Setup Wizard	Guided initial configuration for new assistants
Configuration Flow	Multi-step setup workflow
Platform Settings	General platform preferences

Security & Governance

Feature	Description
Intel SSO	Enterprise single sign-on authentication for all users
Azure Active Directory	Identity provider; user authentication and group management
AGS Entitlements	Fine-grained group-based access for Secure Group sharing
Private by Default	All new assistants default to owner-only access
File Privacy	Assistants with files remain Private (InfoSecurity directive)
Compliance Gate	Required checkbox before Public or Secure Group publishing
Audit / Logs	Telemetry and audit trail for platform operations

> **InfoSecurity Directive:** "Assistants with Files can only be published as Private. This is not a technical restriction, but a direction from InfoSecurity."

External Integrations

Integration	Purpose
Azure Active Directory	Authentication, identity, and group management
Intel SSO	Single sign-on for all Intel employees
AGS Entitlements	Enterprise access group management
Confluence Wiki	Documentation hosting and knowledge base (Learn resources)
LLM APIs	GPT-4, GPT-3.5, and other model API endpoints
Data Pipeline Sources	Automated data ingestion from enterprise systems
InfoSecurity	Policy enforcement for publishing and data compliance

Related Architecture Patterns

iGPT is one of **three complementary AI architecture patterns** at Intel. Each pattern is purpose-built for a distinct class of use cases and maintains its own runtime, identity model, and governance boundaries.

Pattern	Scope	Entry Point
SAP Joule	SAP transactions, navigation, analytics, SAP knowledge	Joule Assistant bar in SAP apps
ECA / Azure Copilot Studio	Enterprise data analytics, agentic workflows, non-SAP integrations	Copilot Agent UX (Teams / MS365)
iGPT Platform (this document)	General-purpose chat, custom assistants, marketplace, document Q&A	igpt.intel.com/chat

High-Level Integration Points

- **iGPT → Joule / ECA**: No direct integration today. iGPT operates as a standalone self-service platform. Users context-switch to Joule for SAP needs or Copilot Studio for ECA data queries.
- **Future Roadmap**: MCP/API integration may enable iGPT assistants to be registered as tools within ECA/Azure agentic workflows, providing a bridge between self-service AI and governed enterprise data.

When to Use iGPT vs Other Patterns

If you need…	Use iGPT	Use Joule or ECA Instead
General-purpose AI chat	✅	—
Custom assistant for your team (no-code)	✅	—
Document Q&A from uploaded files	✅	—
Sharing assistants via Marketplace	✅	—
SAP transaction processing	—	✅ SAP Joule
Enterprise data analytics (Snowflake/Databricks)	—	✅ ECA / Azure
Multi-system agentic workflows	—	✅ ECA / Azure

Selection Guidance

For detailed use case routing, decision frameworks, and guidance on choosing between the three patterns, refer to:

> 📄 **AI Architecture Selection Guide** — `AI-Architecture-Selection-Guide.md`

Related Documents

Document	Location
ECA & AI Enablement Architecture	`ECA_AI_Enablement_Architecture_DocumentV1.md`
SAP Joule Architecture Overview	`Current AI Architecture Options/SAP-Joule-Architecture-Overview.md`
AI Architecture Selection Guide	`AI-Architecture-Selection-Guide.md`

Appendix A: Component Glossary

Component / Abbreviation	Description
iGPT	Intel Generative Pre-trained Transformer – the enterprise AI platform
LLM	Large Language Model – AI models trained on large text corpora (e.g. GPT-4, GPT-3.5)
RAG	Retrieval Augmented Generation – technique that augments LLM prompts with retrieved document context for grounded responses
SPA	Single Page Application – web application architecture that loads a single HTML page and dynamically updates content without full page reloads
SSO	Single Sign-On – one-time authentication granting access to multiple enterprise systems
API	Application Programming Interface – a set of protocols enabling software components to communicate
CRUD	Create, Read, Update, Delete – the four basic operations for data management
UI	User Interface – the visual front-end through which users interact with the platform
DB	Database – structured data storage system
AD	Active Directory – Microsoft's directory service for identity and access management (see Azure AD)
Azure AD	Azure Active Directory – Microsoft cloud-based identity and access management service
AGS	Access Group Service – Intel's enterprise entitlements service for fine-grained group-based permissions
LLM Gateway	Model routing layer that directs requests to the appropriate LLM backend
Vector Database	Stores document embeddings as numerical vectors for semantic similarity search
Assistant Builder	No-code wizard for creating custom AI assistants
Marketplace	Enterprise store for browsing, sharing, and running AI assistants
InfoSecurity	Intel's information security team and policy enforcement function
Confluence	Atlassian wiki platform used for Intel documentation and knowledge bases
GPT-4 / GPT-3.5	OpenAI large language models available via the LLM Gateway
PDF	Portable Document Format – Adobe file format for fixed-layout documents
DOCX	Microsoft Word document format (Office Open XML)
TXT	Plain text file format

Component / Abbreviation	Description
TB / LR	Top-to-Bottom / Left-to-Right – Mermaid flowchart direction indicators used in architecture diagrams

Generated on March 2, 2026
Source: iGPT platform analysis (https://igpt.intel.com/chat)
Modules: 10 | Features: 55 | Security Features: 6 | Integrations: 7

Document Control

Version	Date	Author	Changes
1.0	March 2026	Sajiv Francis	Initial release
1.1	March 2026	Sajiv Francis	Added Related Architecture Patterns section (three-pattern model: Joule, ECA/Azure, iGPT); cross-references to Selection Guide

Classification: Internal Use

Review Cycle: Quarterly

Next Review: June 2026